

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T32 · Linear Graphs $y = mx + c$

Questions

Student book · questions with writing space

■ 19 pages

FAMILY MAP

Linear Graphs $y = mx + c$

F01	Construct a straight-line equation	8 Q 30 marks
------------	---	----------------

F02	Read gradient and intercept structure	8 Q 12 marks
------------	--	----------------

F03	Generate and plot a linear graph	2 Q 10 marks
------------	---	----------------

Working Unit 15

4MA1-1H Q25

UNIT 1
5 marks

25 $PQRS$ is a square.

PR is a diagonal of the square.

P is the point with coordinates $(4, 7)$

R is the point with coordinates $(8, -5)$

Find an equation of the straight line that passes through the points Q and S

Give your answer in the form $ay = bx + c$ where a , b and c are integers.

.....
(Total for Question 25 is 5 marks)

4MA1-2H Q18

2 marks

Find the gradient of a straight line that is perpendicular to $\mathbf{P}$

.....

Question 21

4MA1-2H Q9 whole

MODULAR UNIT 1

2 marks

- 9 A straight line passes through the points with coordinates $(0, -3)$ and $(2, 0)$

Find an equation of the line.

.....

4MA1-1H Q13

3 marks

Find an equation of the straight line that passes through the points A and B

Question 25

4MA1-2H Q18 M01

MODULAR UNIT 1

4 marks

- 18 The straight line **L** is perpendicular to the straight line with equation $2x + y = 9$ and passes through the point with coordinates (8, 11)

Find an equation for **L**

Give your answer in the form $y = mx + c$

.....

Question 08

4MA1-1H Q10 Q10(a)

MODULAR UNIT 1

2 marks

10 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

(2)

Question 09

4MA1-1H Q10 Q10(b)

MODULAR UNIT 1

1 marks

10 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

(b) Find the coordinates of the point where **L** crosses the y -axis.

(.....,)
(1)

(b) Write down the gradient of line **L**.

(c) Does the point with coordinates $(100, -302)$ lie on line **L**?
You must give a reason for your answer.

19 (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

(a) Work out the gradient of AB .

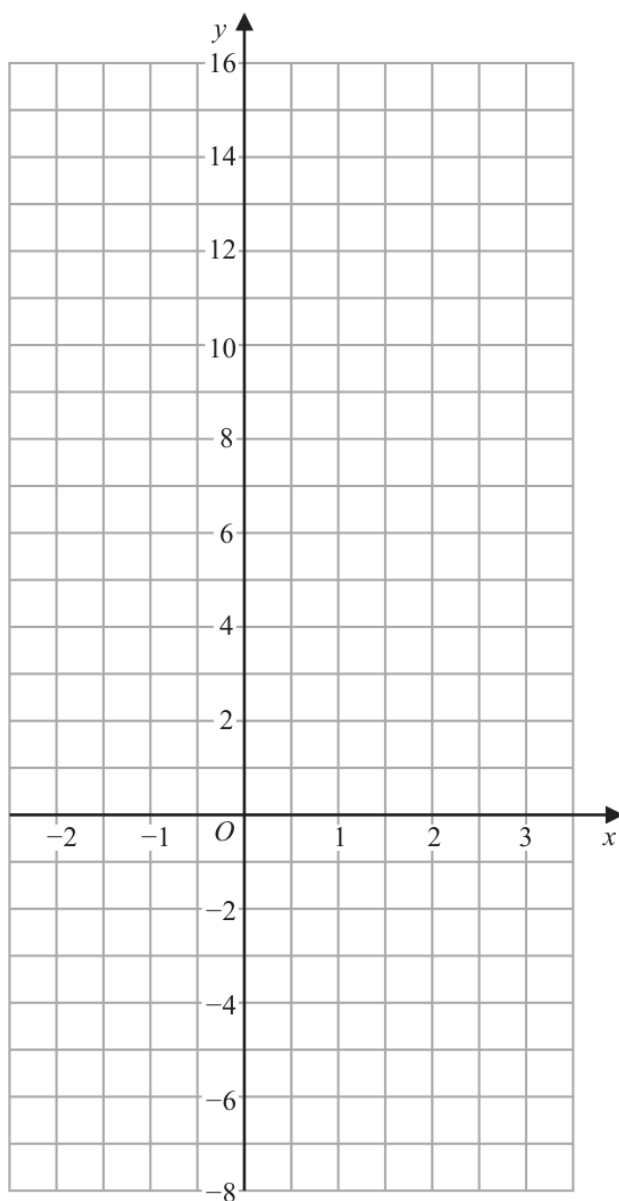
(b) Write down the gradient of a straight line that is perpendicular to $\mathbf{L}$.

The straight line **M** is perpendicular to **L** and intersects the y -axis at the point $(0, 8)$

(.....,)

Calculate the area of triangle ABC .

- 3 On the grid, draw the graph of $y = 7 - 4x$ for values of x from -2 to 3



Find an equation for **L**.

- (1)

(2)

(2)

- (d) Write down an equation of the line.